

INDU 211 · Rapid Review Sheet (1-Page Cheatsheet)

Part 1: Chapters 1 & 2 — Foundations of Industrial & Systems Engineering

1. Etymology & Foundational Duality

- **Root:** Latin *ingenium* = inborn talent, cleverness, creative inventiveness → Engineer = inventive problem-solver.
- **Science vs. Engineering:**
 - **Science:** Quest for *basic knowledge* | Method: Observe → Hypothesize → Controlled Experiment | Output: Theories.
 - **Engineering:** Application of science to solve human problems | Goal: *A better life* | Output: Functional systems.
 - **Interplay:** Science feeds discoveries to engineering; engineering feeds problems back to science. Math is the common foundation.

2. Chronology of Pioneers (Industrial Revolution → Scientific Management)

- **Charles Babbage** (1800s): **Division of Labor**; specialized tasks reduce wage/training costs; harmonious labor relations.
- **Frederick W. Taylor** (1900s): "**Scientific Management**"; stopwatch time studies; standardization of tools & work methods.
- **Henry Ford** (1913): **Mass Production & Moving Assembly Line**; interchangeable parts; simple jobs for minimally trained workers.
- **Frank & Lillian Gilbreth** (1910s): **Motion Analysis & Human Factors**; eliminated wasted motions (therbligs); ergonomics, fatigue & psychology.
- **Henry L. Gantt** (1917): **Gantt Chart**; visual milestone bar chart comparing planned vs. actual project schedules.
- **Walter Shewhart** (1920s): **Statistical Quality Control (SQC)**; control charts separating random noise from assignable defects.
- **Operations Research (WWII)**: Applying mathematical modeling, optimization & linear programming to operational logistics → Big Data & AI.

3. Professional Regulation & Ethics (Quebec & Canada)

- **Regulatory Hierarchy:** **Engineers Canada** (National body) → **CEAB** (Accredits B.Eng degrees) → **OIQ** (Quebec provincial regulator, ~55k P.Engs).
- **Legal Definition ("Practice of Engineering"):** Any act of planning, designing, composing, evaluating, advising, reporting, directing, supervising, or managing work requiring engineering principles and concerning the safeguarding of **life, health, property, economic interests, public welfare, or the environment**.
- **Ethical Mandate:** When company cost pressures conflict with public safety, the engineer's primary legal & ethical obligation is **ALWAYS to public safety**.

4. Systems Theory & The Production Model

- **System:** A set of interrelated components acting together toward a common objective.
- **Open-Loop:** No feedback, cannot self-correct ("*car without a driver*").
- **Closed-Loop:** Monitors output, compares against standard, applies feedback control ("*car with a driver*").
- **Universal Production Loop:**

Inputs (Capital, Materials, Labor, Market) → [Conversion Process] → Outputs (Goods/Services)

↑ ————— Corrective Action ← Performance Monitoring / Feedback ————— ↓

5. The Decision-Making Hierarchy

Level	Time Horizon	Key Questions	Typical Industrial Examples
Strategic	Long-term (3 – 10 yrs)	What, How, Where?	New product lines, plant locations, capital investments.
Tactical	Medium-term (1 – 12 mos)	How much, When?	Aggregate production schedules, inventory safety stock, seasonal hiring.
Control	Short-term (Daily/Hourly)	Keep line running!	Shift worker dispatching, machine repair, tracking scrap/defects.

6. The IE Identity & Practice

- **Core Motto:** Work with **PEOPLE** to make systems **BETTER, FASTER, SAFER, CHEAPER**.
- **Two Design Levels:**
 1. **Human Activity Systems:** Physical workplace (layout, ergonomics, methods, safety, material handling).
 2. **Management Control Systems:** Policies & metrics (forecasting, MRP, inventory, quality control, scheduling).
- **Human Factor (Dale Carnegie):** **15%** financial success is technical knowledge; **85%** is human engineering (leadership, communication, empathy).
- **IE in the Age of AI:** Enhances IE; enables predictive maintenance, real-time bottleneck detection, and digital twin simulation.